

SMART FURNITURE INVENTORY ANALYSIS

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PROJECT OBJECTIVE

The objective of this project is to convert raw furniture inventory data into a structured operational inventory analysis system using Excel-based reporting, stock monitoring, dashboard visualization, and automation-oriented operational insights.

PROJECT MODULES COMPLETED

1. Raw Inventory Data Creation

The furniture inventory PDF data was manually converted into a structured Excel database format.

The raw inventory sheet contains:

- Item ID
- Category
- Subcategory
- Item Name
- Stock Quantity
- Stock Status
- Priority Level
- Remarks

Purpose:

To create a clean and structured inventory database for operational analysis and future scalability.

2. Stock Status Automation

Excel formulas were used to automatically classify inventory stock conditions into:

- Critical
- Medium

- Healthy

Automation logic was applied based on stock quantity ranges.

Example:

- Low stock → Critical
- Medium stock → Medium
- High stock → Healthy

Purpose:

To simplify inventory monitoring and reduce manual operational tracking.

3. Conditional Formatting & Visual Indicators

Conditional formatting was implemented to visually highlight stock conditions.

Color Logic Used:

- Red → Critical
- Yellow → Medium
- Green → Healthy

Purpose:

To improve quick inventory understanding and operational visibility.

4. Stock Health Analysis

A dedicated stock health analysis sheet was created containing:

- Total Inventory Summary
- Critical Stock Count
- Medium Stock Count
- Healthy Stock Count
- Inventory Condition Summary
- Key Operational Insights
- Critical Stock Item Tracking

Excel formulas used:

- COUNTA

- COUNTIF
- Stock classification formulas

Purpose:

To create operational intelligence from raw inventory data.

5. Dashboard Development

A structured inventory dashboard was created to provide quick operational understanding without reading long inventory tables.

Dashboard sections include:

- KPI Inventory Overview
- Category Distribution
- Critical Stock Items
- High Availability Inventory

Purpose:

To help management quickly understand inventory health, risk levels, operational readiness, and inventory distribution.

6. Automation & AI Improvement Ideas

Future operational improvement ideas were proposed, including:

- Automatic low-stock alerts
- WhatsApp inventory notifications
- QR code inventory tracking
- Barcode inventory system
- AI-based stock prediction
- Automated operational reporting
- Digital inventory management improvements

Purpose:

To explore future operational automation opportunities and improve inventory efficiency.

PROJECT OUTCOME

The project successfully transformed raw inventory data into a structured operational inventory analysis and reporting system using Excel-based automation and business analysis concepts.

The system improves:

- Inventory visibility
- Operational understanding
- Risk identification
- Stock monitoring
- Dashboard reporting
- Future automation readiness

TOOLS & CONCEPTS USED

- Microsoft Excel
- Excel Formulas
- Conditional Formatting
- Inventory Classification
- Operational Analysis
- Dashboard Structuring
- Automation Planning
- AI-Oriented Improvement Ideas

FINAL SUMMARY

This project demonstrates a structured approach toward operational inventory management, inventory intelligence, automation thinking, and business process improvement using Excel-based analytical systems.